Recruitment Platform

Milestone 1: System Definition and Architecturally Significant Requirements

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Course: Principles of Software Architecture

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Contribution Table

The table below shows the work each team member did for this milestone.

Member	Student ID	Contribution
Mahmoud Ahmed	231014373	Quality attribute scenarios
Ahmed Farouk	231014395	Constraints, Architecturally significant requirements (ASRs)
Abdullah Emad Abdelmoniem	231006822	System overview, Stakeholders and concerns
Hanin Maged Gamal	231014560	Functional requirements
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1. System Overview

Hiring inside a company is split across people who do not share the same work or the same need to see data. Human resources keeps candidate profiles, job openings, and the movement of each application through the hiring stages. Interviewers need only the applications assigned to them, and they record an evaluation when the interview is done. An administrator creates the staff accounts, sets roles, and can enable or disable access. Candidates do not sign in. The platform holds their profiles, résumés, and evaluations, so who may see that information is part of the design. Today that work is easy to lose in mail, spreadsheets, and shared folders: a stage change is not tied to who made it, an interviewer can be shown applications that are not theirs, and a résumé sits apart from the person it belongs to. The proposed system is an internal recruitment platform that keeps candidates, jobs, applications, stage changes, interviewer assignments, and evaluations in one hiring process, with each role seeing only the work it is allowed to do.

The platform is architecturally interesting because the shape of the system is driven by quality attributes and constraints. Staff accounts should be authenticated against the company's existing directory (LDAP), so identity is not owned entirely inside the application. Candidate records, hiring-pipeline records, and account records change on different schedules and are used by different roles, so they are proposed as separately deployable services behind a web frontend: an authentication service, a candidate service, and an application and hiring-pipeline service. That split supports modifiability — a change to evaluations need not be released with a change to login — and availability, because one service can keep serving while another is down. It also creates the tension the architecture has to live with. A hiring step that needs a user, a candidate, and an application must call more than one service, so it is slower than the same step inside a single application. A change that touches all three is not one immediate update: each service stores its own data, and the others learn of it by reference rather than by a shared transaction. The architecture therefore has to decide, up front, where consistency may lag and where a cross-service call is acceptable.

2. Stakeholders and Concerns

Stakeholder	Key concerns
HR specialist	Can create and update candidates and job openings, move an application through the allowed stages, and assign an interviewer without exposing the rest of the pipeline to people who are not on that hire. Needs the current stage and assignment to be trustworthy enough to act on during a working day.
Interviewer	Sees only applications assigned to them, and can submit an evaluation for those applications. Must not browse other candidates, other jobs, or hiring decisions outside that assignment.
Administrator	Registers HR and interviewer accounts, including accounts linked to the company directory, and can change role and account status. Also needs a path to correct candidates, jobs, and applications when HR cannot. Concerned that a disabled account stops being accepted quickly.

Stakeholder	Key concerns
Business owner	The hiring process stays inside the company, stage history can be explained after the fact, and the platform can be adopted without replacing the existing staff directory. Cares that time-to-hire is not slowed by a design that calls many services for every ordinary step.
Candidate	Does not use the platform, but the profile, résumé, and evaluations are about them. Cares that only the staff involved in that hire can see those records, and that the information HR keeps is accurate enough to be used in a hiring decision.
Development team	Each service can be changed, tested, and released on its own. Contracts between services stay explicit, because a candidate identifier or a user identifier stored in another service is a reference, not a database foreign key. Wants a clear rule for what happens when a called service is unavailable.

3. System Context

Figure 1 shows the Recruitment Platform as a single software system, the three staff roles that use it, and the external LDAP directory it uses to authenticate staff credentials. The diagram source file (C4_level_1.drawio) is included in the repository.

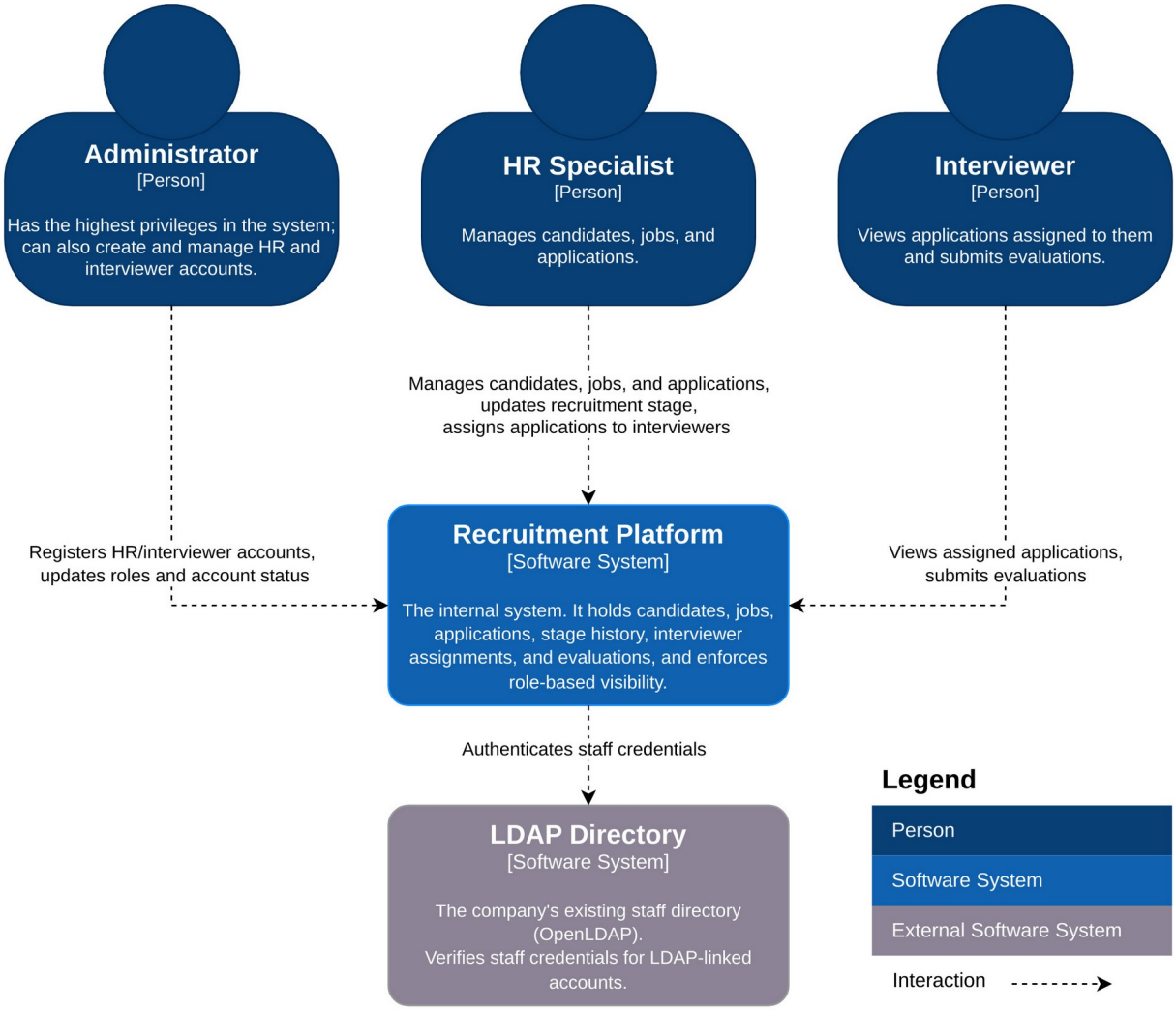


Figure 1: System Context Diagram (C4 Level 1) – Recruitment Platform
*Note: Candidates don't use the system; they provide their CV to HR specialists.

4. Functional Requirements

These requirements describe the main functions of the Recruitment Platform. Each row has a MoSCoW priority and shows whether the Milestone 3 prototype will include it. Candidates have no accounts.

ID	Use case	Functional requirement	Priority	M3 prototype
FR-01	Log in and log out	An Admin, HR user, or Interviewer shall be able to log in with valid credentials, including credentials checked against the company directory, and shall be able to log out so the session can no longer be continued.	Must Have	Yes
FR-02	Register team members	An Admin shall be able to register an HR user or an Interviewer, assign that role, and link the account to the company directory when the person already has a directory account.	Must Have	Yes
FR-03	Refresh a session	A signed-in user shall be able to continue the session with a still-valid refresh token after the access period ends.	Should Have	No
FR-04	Change role and account status	An Admin shall be able to change a team member's role and disable or enable the account. A disabled account shall no longer be accepted at login.	Should Have	No
FR-05	Manage candidates	An HR user or an Admin shall be able to create, view, and update candidate records.	Must Have	Yes
FR-06	Upload and parse a CV	An HR user or an Admin shall be able to upload a candidate's CV and have the platform parse it into the candidate record.	Must Have	Yes
FR-07	Manage jobs	An HR user or an Admin shall be able to create, view, and update job records, and publish a job when it is ready for applications.	Must Have	Yes
FR-08	Manage applications	An HR user or an Admin shall be able to create and view an application that links one candidate to one job.	Must Have	Yes
FR-09	Update recruitment stage	An HR user shall be able to move an application to another stage only when that move is allowed. The stage history shall record who made the change.	Must Have	Yes
FR-10	Assign an interviewer	An HR user shall be able to assign an interviewer, chosen from the registered interviewers, to an application that is in screening or interview.	Must Have	Yes
FR-11	View assignments and submit an evaluation	An Interviewer shall be able to view only the applications assigned to them, and submit an evaluation for an assigned application that is in screening or interview. An HR user shall be able to view all evaluations submitted for an application.	Must Have	Yes

ID	Use case	Functional requirement	Priority	M3 prototype
FR-12	Filter lists	An HR user shall be able to filter candidates by tag, jobs by publication status or tag, and applications by job, candidate, or stage. Each list can still be opened in full without a filter.	Could Have	No
FR-13	Email notifications	The platform shall send an email to an interviewer when they are assigned to an application.	Won’t Have	No

MoSCoW priorities

Priority	Meaning
Must Have	Essential for the main recruitment workflow.
Should Have	Important, but less essential than the core functions.
Could Have	Useful if time allows.
Won’t Have	Not planned for this milestone.

5. Quality Attribute Scenarios

Each scenario uses the six-part form. Together they cover five quality attributes: security, consistency, modifiability, performance and availability.

Security

QAS-01: CV file validation

Source	HR specialist
Stimulus	Uploads a candidate CV that is not a real PDF or Word file (for example, an executable or a script renamed to .pdf)
Artifact	candidate-service
Environment	Normal operation
Response	The service checks the actual file type and size before saving. Files that are not PDF or DOCX, or are larger than 5 MB, are rejected and the HR user is told why.
Response measure	100% of files that are not PDF/DOCX or are over 5 MB are rejected within 1 second and are never stored.

QAS-02: Password guessing

Source	Unauthorized user
Stimulus	Repeatedly tries to log in with guessed or leaked credentials of an HR specialist or interviewer
Artifact	auth-service
Environment	Normal operation
Response	After 5 failed attempts within 10 minutes, the account is temporarily locked and the attempts are logged so the Admin can see them.
Response measure	100% of accounts are locked for 15 minutes after the 5th failed attempt, and every failed attempt is recorded with time and username.

QAS-03: Role-based visibility

Source	Interviewer
Stimulus	Requests the details or CV of an application that is not assigned to them
Artifact	application-service, candidate-service
Environment	Normal operation
Response	The request is refused and no candidate data is returned.
Response measure	100% of requests for applications not assigned to the interviewer are refused, on every service.

Consistency

QAS-04: Cross-service references

Source	HR specialist
Stimulus	Creates an application or assigns an interviewer using a candidate or interviewer that was deleted or disabled
Artifact	application-service
Environment	Normal operation
Response	Before saving, application-service checks the candidate with candidate-service and the interviewer with auth-service. If the record is missing or disabled, the request is rejected with a clear message.
Response measure	0 applications or assignments are saved that point to a missing or disabled candidate or interviewer.

Modifiability

QAS-05: Adding notifications

Source	Product owner
Stimulus	Asks for an email notification when an interviewer is assigned or an application changes stage
Artifact	application-service
Environment	Development time
Response	application-service publishes events (for example, ApplicationAssigned) to a message broker. A new notification-service reads these events and sends the emails.
Response measure	The notification-service is added in at most 3 person-days. application-service only adds event publishing, no existing API changes, and auth-service and candidate-service are not changed or redeployed.

Performance

QAS-06: CV uploads under load

Source	HR specialists
Stimulus	50 HR users upload CV PDF files (up to 5 MB each) at the same time
Artifact	candidate-service
Environment	Peak load
Response	candidate-service saves the file and responds right away; parsing the CV into the candidate record happens in the background.
Response measure	95% of uploads are acknowledged in under 2 seconds, 99% succeed with no dropped connections or HTTP 500 errors, and parsed data appears on the candidate record within 30 seconds.

Availability

QAS-07: Service crash

Source	Infrastructure
Stimulus	candidate-service or its MySQL database container crashes unexpectedly
Artifact	candidate-service and its database
Environment	Normal operation
Response	Docker health checks detect the failure and restart the container, and the service reconnects to the database automatically. While it is down, auth-service and application-service keep working, and requests that need candidate data return a clear error instead of hanging.
Response measure	candidate-service is back within 30 seconds; requests that need it fail within 2 seconds while it is down; auth-service and application-service have no downtime.

6. Constraints

The following constraints limit the choices we can make when designing and building the Recruitment Platform.

6.1 Technical constraints

ID	Constraint
C-01	Staff must log in using the company's existing LDAP directory.
C-02	The system is split into separate parts that can be deployed on their own: an authentication service, a candidate service, an application and hiring-pipeline service, and a web frontend.
C-03	Each service has its own database. When one service needs a record from another service, it stores only its ID (no shared database and no foreign keys between services).
C-04	The backend services must be written in Java.

6.2 Business constraints

ID	Constraint
C-05	The platform is internal. Only company staff (Admin, HR, Interviewer) use it, and candidates do not have accounts.
C-06	The platform must work with the company's current staff directory, not replace it.
C-07	Candidate profiles, CVs and evaluations can only be seen by the staff working on that hire.

6.3 Organizational constraints

ID	Constraint
C-08	The team has five students.
C-09	The project has four milestones with fixed deadlines. Milestone 1 is due on 23 October 2026, and late work loses 10% per day.
C-10	All project work, including the source files of the diagrams, is submitted through GitHub.
C-11	The Milestone 3 prototype will only include the requirements marked "Yes" in Section 4.

7. Architecturally Significant Requirements (ASRs)

We ranked the requirements from Sections 4 to 6 by two things: how important they are to the business, and how much they affect the architecture (H = high, M = medium, L = low). A requirement is an ASR when both are high, or when getting it wrong would mean changing the structure of the system instead of changing one module.

7.1 Prioritized table

Rank	Requirement(s)	Business value	Arch. impact	Result
1	Access control: QAS-02, QAS-03, FR-01, FR-02, FR-04, FR-11, C-01, C-07	H	H	ASR-1
2	Consistency between services: QAS-04, FR-08, FR-09, FR-10, C-02, C-03	H	H	ASR-2
3	Availability when one service fails: QAS-07, C-02	H	H	ASR-3
4	CV upload and parsing: QAS-06, QAS-01, FR-06	H	M	ASR-4
5	Adding new features without touching other services: QAS-05, FR-13, C-02	M	H	ASR-5
6	FR-03 Refresh a session	M	M	–
7	FR-05 Manage candidates, FR-07 Manage jobs	H	L	–
8	FR-12 Filter lists	L	L	–
–	C-04 to C-06, C-08 to C-11	–	–	Fixed limits, not ranked

FR-05 and FR-07 are important to HR, but they are simple create/view/update functions that any structure can support, so their architectural impact is low.

7.2 Utility tree

Quality attribute	Refinement	Scenario	(BV, AI)
Security	Access control	QAS-03: an interviewer is refused 100% of the time when opening an application not assigned to them	(H, H)
Security	Authentication	QAS-02: account locked for 15 min after 5 failed logins	(H, H)
Security	File safety	QAS-01: non-PDF/DOCX or oversized CVs rejected within 1 s	(H, M)
Consistency	Cross-service references	QAS-04: 0 applications or assignments pointing to a missing or disabled record	(H, H)
Availability	Partial failure	QAS-07: candidate-service back within 30 s; other services stay up	(H, H)

Quality attribute	Refinement	Scenario	(BV, AI)
Performance	CV upload under load	QAS-06: 95% of uploads acknowledged in under 2 s with 50 users	(H, M)
Modifiability	New service	QAS-05: notification-service added in at most 3 person-days without changing other services	(M, H)

BV = business value, AI = architectural impact.

7.3 Why these are the top 5

ASR-1: Access control (H, H)

The System Overview says that today "an interviewer can be shown applications that are not theirs", and the candidates' main concern is that only staff on their hire can see their records (C-07). The rule is not only about the role, it also depends on data: an interviewer may only see applications assigned to them, and the assignments are stored in application-service, not in auth-service. Login also has two paths, local accounts and LDAP (C-01), and a disabled account must stop working (FR-04). This forces us to decide where checks happen: auth-service issues a token after login, every service checks the token and role on its own, and application-service also checks assignment ownership. Hiding buttons in the frontend is not enough to meet QAS-03.

ASR-2: Consistency between services (H, H)

Because each service has its own database (C-03), an application only stores the ID of a candidate and the ID of an interviewer. FR-08, FR-09 and FR-10 all depend on records owned by other services. Without a rule, an application can point to a candidate that was deleted or an interviewer that was disabled. QAS-04 makes application-service check these records before saving, and records should be disabled instead of deleted so the stage history in FR-09 still makes sense later.

ASR-3: Availability when one service fails (H, H)

The main reason for splitting the system into services (C-02) is that one service can keep working while another is down. QAS-07 only holds if every service handles a failed call: timeouts, a clear error, and no waiting forever. This conflicts with ASR-2: if application-service must check candidate-service before every save, it cannot create applications while candidate-service is down. This availability versus consistency trade-off is the one named in our project idea, and we will decide it in Milestone 2.

ASR-4: CV upload and parsing (H, M)

FR-06 is where every hire starts, and parsing a PDF is the heaviest work in the system. If parsing happens inside the upload request, the upload becomes slow and QAS-06 fails under load. This pushes us to save the file first and parse it in the background, and to keep files in storage that is not tied to one running copy of candidate-service. QAS-01 adds file type and size checks before anything is saved.

ASR-5: Adding features without touching other services (M, H)

Users will not notice this directly, so its business value is medium, but it affects how services talk to each other. QAS-05 shows that if application-service publishes events, a new notification-service (FR-13) can be

added without changing the other services. The cost is performance and complexity: a message broker is one more part to run, and any hiring step that needs data from several services is slower than in a single application. This is the second trade-off in our project idea, performance versus modifiability.

7.4 Trade-offs to resolve in Milestone 2

Trade-off	Between	What decides it
T1	Availability (ASR-3) vs. consistency (ASR-2)	Checking other services before every save vs. keeping a local copy of the data
T2	Modifiability (ASR-5) vs. performance (ASR-4)	How many calls between services one hiring step needs, and whether events go through a message broker
T3	Security (ASR-1) vs. usability	How long a login stays valid, and the number of failed attempts before lockout

References

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